It was a blazing sunny day and the air was gritty. He was a small kid looking tired and faint, I offered him water and it turned out that he had a kidney problem due to contaminated drinking water. Pondering for plausible solution, that day I decided to work for energy and water crises. So, I moved from physics major to energy program in my Master studies at National University of Science and Technology, (NUST), Islamabad Pakistan. While I majored in physics, I opted an elective course of introductory renewable energy technologies and audited a MOOC on edX, ‘Solar Energy’ taught by Dr. Arno Smets in TU Delft. It broadened my erudition perspective and strengthened my concepts in this field so I steered my research journey towards solar energy.

Though I grew up in a male dominated society and lack elder brother or anyone to guide, I am an independent fanatic about my life goals. I strived for my life on my own when my father was abroad and I had to cope my studies with domestic duties independently. My fervor to equip myself with versatile skills would never let me rely merely on the school’s curriculum and so to retort my inquisitiveness, I used to hunt different MOOCs, would partake in seminars and conferences, and enroll in various skills training programs. Eagerness to be a pragmatic adroit affected my grades gravely but I used to believe my future lies beyond theoretical exams.

My prowess stimulated me to serve the society potentially. Even during earlier stage of my academic career, when I had just done with my intermediate exams, I initiated an academy of spoken English and basic computing skills for females in my community to furnish their communication skills. That initiative was successful beyond my expectations and most of those females are now supporting a house hold.

I proposed and presented an idea in a ‘Hult prize business competition, 2014’, which was selected among top 11 out of more than 200 entries throughout Pakistan. That bequeathed me confidence that I retain good leadership qualities. Later on I participated in many conferences, workshops, business competitions and research showcase platforms. I was selected for a ‘Training of trainers program for young leaders’ by Faces Pakistan and now I am a well-accomplished mentor and facilitating individuals by organizing peace building and leadership training workshops.

It has been a life time achievement, to study in the country’s top university and winning highly competitive USAID merit scholarship, as a part of which I visited Oregon State University, USA as an exchange scholar to carry out advanced research in energy. Working on hydro power project in progressive labs, research design and methodology seminars, communication developing sessions and interdisciplinary workshops motivated me to apply for Ph.D. in USA. After coming back to Pakistan, I volunteered public awareness seminars in Department of Physics and Department of Environmental Sciences at University of Haripur to promote use of renewable energies. Also I directed students for their research and motivated them to work for mitigating energy crisis. I was able to develop interest of students in solar technologies, carbon capture technologies and climate engineering to reduce the effects of global warming.

My study objective is to pursue a doctorate degree in a field related to sustainable energy and contribute for the solution of energy and water crisis. As my graduate research during MS in Energy System Engineering at NUST, I developed a small scale solar humidification dehumidification (HDH) water desalination system and evaluated its thermal behavior. I was able to improve the system performance by enhancing the humidification rate and published my research in IEEE international conference. My article on a similar system is under review in an Elsevier Journal.

I was awarded teaching and research assistantship during my MS studies and I worked on heat transfer enhancement by nano fluids, synthesis of Titania nano particles and stability analysis, theoretical modelling of solar thermal power plants, and synthesis and fabrication of dye sensitized solar cells. I also assisted my professor in teaching the Solar Engineering course and Research Methodology.

This MS program was a good start for my career in energy as it was a complete package, covering wide range of energy related courses (including solar energy, wind energy, bio fuels, hydropower and geothermal energy), organizing international conferences, seminars and interdisciplinary workshops including industrial visits for theoretical as well as practical understanding, semester projects to gain research skills and providing a platform to enhance communication, management and presentation skills. After completion of my coursework, I also audited additional courses related to solar engineering and visited Pakistan Council of Renewable Energy Technologies (PCRET) and Pakistan Council of Research in Water Resources (PCRWR) to get insight of installed systems there and explore research facilities.

Being an active student, I had organized the departmental sports gala as a female representative, won two medals, joined the university gymnasium to play badminton in evening and also maintained the habit of morning walk. So I organized my time table and kept a balance between studies and extra-curricular. That gave me the spirit and energy to carry out my research without taking stress. I also love hiking because that teaches me diligence, determination and a passion to move forward, keeping my spirits high and boosting my motivation to meet my life goals.

I'm looking for a Ph.D. program where I would take coursework related to materials and solar engineering to get more prowess for further advanced research. I want to work on synthesis of new material for solar energy applications.

I'm passionate to do something great using my skills in the field of solar energy. I have learnt from the top ranked University in Pakistan and grasped the analytical skills and basic research skills from top ranked research institutes in USA. I believe my previous research is a good fit for carrying out further research in solar energy.